KM-02-KT05: Cartesian Coordinate System

Written Assignment Submission Template

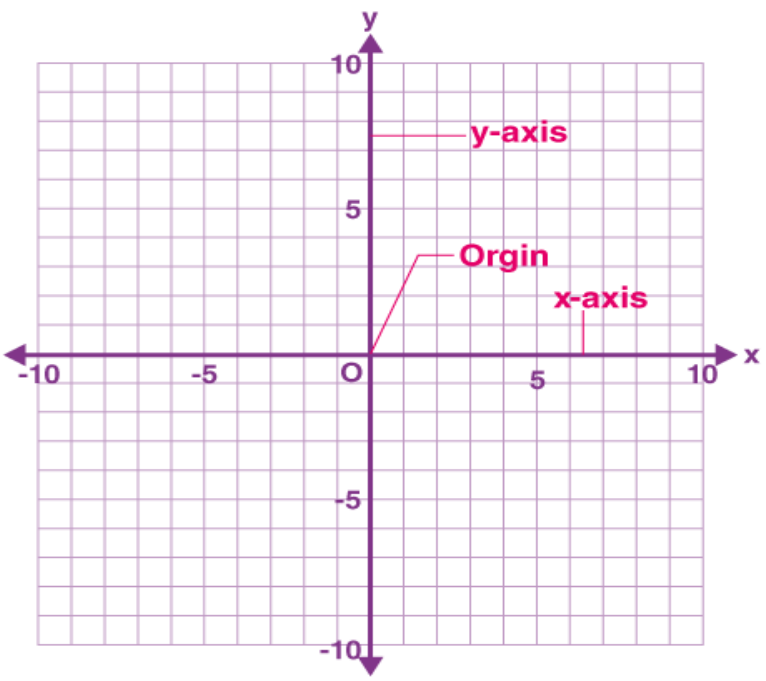
Title: Understanding the Cartesian Coordinate System
Due Date: 23 October 2025
Instructions: Complete all four sections below. Submit your responses as a single document via the LMS. Use full sentences and ensure your work is clear, structured, and original.

Section 1: Definitions and Terminology. 8 Marks

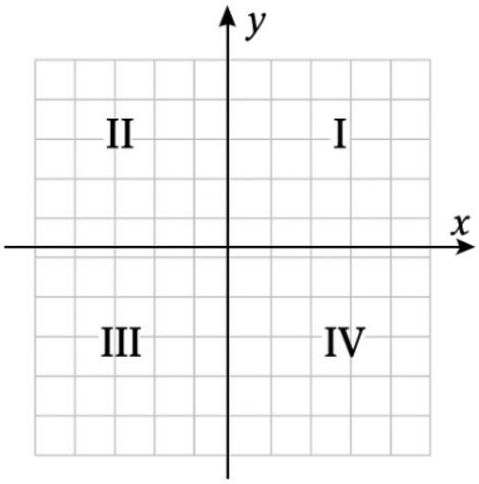
1. **Define the Cartesian coordinate system in your own words:**

The Cartesian coordinate system is a two-dimensional plane defined by two perpendicular number lines that intersect at a zero point. It provides a method to precisely locate any point using a pair of numerical values (coordinates), which represent its horizontal and vertical positions.

2. **Draw a basic Cartesian plane as per the example, and identify and define the following terms on your diagram:**
   - Origin
   - x-axis
   - y-axis
   - Coordinates
   - Quadrant



- **Origin:** The point where the x-axis and y-axis intersect. It is the reference point for the entire system and has coordinates (0, 0).
- **x-axis:** The horizontal number line in the coordinate system. Positions to the right of the origin are positive, and positions to the left are negative.
- **y-axis:** The vertical number line in the coordinate system. Positions above the origin are positive, and positions below are negative.
- **Coordinates:** An ordered pair of numbers (x, y) that defines the location of a point. The first number (x) indicates the horizontal position from the origin, and the second number (y) indicates the vertical position.
- **Quadrant:** Any of the four regions created by the intersection of the x-axis and y-axis. The quadrants are numbered counterclockwise from the top-right (I) to the bottom-right (IV).



Section 2: Quadrant Identification. 8 Marks

Instructions: Plot points A–E on your diagram. Then complete the table below. Use Roman numerals I–IV for quadrant identification. Explain your reasoning based on the signs of x and y.

Point	Coordinates	Quadrant	Reasoning
A	(4, 5)	**Quadrant I**.	Both the x-value (4) and y-value (5) are positive.
B	(–3, 2)	**Quadrant II**.	The x-value (-3) is negative, and the y-value (2) is positive.
C	(–2, –6)	**Quadrant III**.	Both the x-value (-2) and y-value (-6) are negative.
D	(5, –1)	**Quadrant IV**.	The x-value (5) is positive, and the y-value (-1) is negative.
E	(0, –3)	**Does not lie in a quadrant**	Its x-value is 0, meaning it lies directly on the y-axis. Points on the axes are not considered to be in any quadrant.

Section 3: Coordinate Plane Interpretation. 8 Marks

Instructions: Study the pattern formed by points A to D. Then respond to the following:
1. What do you notice about the distribution of points across the quadrants?
2. How does the coordinate plane help us organise and interpret this information?
3. Point E lies on the y-axis. What does this tell you about its x-value—and why is this

significant?

1. **1.What do you notice about the distribution of points across the quadrants?**
   The points are distributed across all four quadrants, with each point A-D representing a unique combination of positive and negative sign conventions for its x and y coordinates.
2. **2. How does the coordinate plane help us organize and interpret this information?**
   The coordinate plane provides an immediate visual framework for understanding the relationship between points. By simply looking at a point's quadrant, we can instantly infer the signs of its coordinates, which allows us to categorize and compare data points efficiently.
3. **3. Point E lies on the y-axis. What does this tell you about its x-value—and why is this significant?**
   Point E lying on the y-axis tells us that its x-value is exactly 0. This is significant because it means the point's position is independent of any horizontal movement from the origin. In a data context, this could represent a scenario where the variable measured on the x-axis has no influence or a neutral value.

Section 4: Plotting and Interpretation. 16 Marks

Dataset:

Phase	Hours Spent
Ingestion	12
Validation	18
Transformation	14
Integration	20

Instructions:

1. Plot the data on a new Cartesian coordinate plane using:
   - **x-axis**: Project phases (1 = Ingestion, 2 = Validation, 3 = Transformation, 4 = Integration)
   - **y-axis:** Hours spent on data cleaning
2. Label each point clearly and indicate which quadrant(s) the data falls into.
3. Interpret the graph in 100–150 words:

```
20 +                    D (4, 20)
   |
   |
18 |        B (2, 18)
   |
   |
16 |
   |
14 |              C (3, 14)
   |
   |
12 +   A (1, 12)
   |
   |
   --+-----+--------+----+-- X-Axis (Project Phase)
     1    2    3    4
   Ing Val Tra Int
```

Interpretation Response:

The graph reveals a fluctuating but overall increasing trend in the time spent on data cleaning across project phases. The workload starts at 12 hours (Ingestion), peaks at 18 hours (Validation), dips slightly to 14 hours (Transformation), and then reaches its highest point at 20 hours (Integration). The points do not follow a straight line, indicating that the effort required is not constant and is likely dependent on the specific tasks of each phase. The upward movement from Phase 3 to Phase 4 is the most significant, suggesting Integration is the most labour-intensive phase for data cleaning.

Marking Rubric

Section	Criteria	Excellent (5)	Competent (3–4)	Needs Improvement (1–2)
1	**Terminology & Definitions**	Clear, accurate, learner-friendly	Mostly accurate; minor gaps	Vague or incorrect definitions
	Use of Key Terms	All terms defined with precision and relevance	Most terms defined; some lack clarity	Several terms missing or unclear
2	**Quadrant Identification**	All coordinates correctly placed and explained	Mostly correct; some reasoning unclear	Multiple errors or missing explanations
	Reasoning & Clarity	Clear explanation using signs and quadrant logic	Basic reasoning; some gaps	Reasoning unclear or absent
3	**Graph Interpretation**	Thoughtful, pattern-based analysis	Basic interpretation; limited insight	Misinterprets or omits key trends
	Use of Coordinate Concepts	Coordinates used to support interpretation	Some use of coordinate ideas	Little or incorrect use of coordinate ideas
4	**Plot Accuracy**	All points correctly plotted and labelled	Minor errors in plotting or labelling	Multiple errors or missing points
	Interpretation of Trends	Clear, insightful analysis of workload and progression	Basic interpretation; limited insight	Misinterprets or omits key trends
	Use of Terminology	Accurate use of terms (e.g., quadrant, axis, coordinates)	Some terminology used correctly	Little or incorrect use of terminology

Section	Criteria	Excellent (5)	Competent (3–4)	Needs Improvement (1–2)
	Clarity & Structure	Well-organised, concise, and easy to follow	Mostly clear; minor issues with flow	Disorganised or difficult to follow

Total: 40 Marks